

Practical-10

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himasree@LAPTOP-41COT5AA:~/OSLab$ nano redirect.c
himasree@LAPTOP-41COT5AA:~/OSLab$ nano input.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc -Wall -g redirect.c -o redirect
himasree@LAPTOP-41COT5AA:~/OSLab$ ./redirect
Opening input.txt...
himasree@LAPTOP-41COT5AA:~/OSLab$ dup2(output_fd, STDOUT_FILENO);
-bash: syntax error near unexpected token `output_fd,'
himasree@LAPTOP-41COT5AA:~/OSLab$ cat output.txt
=== I/O REDIRECTION DEMO ===
Standard input is now redirected from input.txt.
Standard output is now redirected to output.txt.

Contents of input.txt:
Hello from the input file!

This text will be read through standard input.
The dup2() system call redirects stdin from the keyboard
to this file.

Redirection completed successfully.
himasree@LAPTOP-41COT5AA:~/OSLab$ echo "This is my inode experiment file." > original.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ ls -li original.txt
11654 original.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ ln original.txt hardlink.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ echo "Added through hard link." >> hardlink.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ cat original.txt
This is my inode experiment file.
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Added through hard link.
himasree@LAPTOP-41COT5AA:~/OSLab$ ln -s original.txt symlink.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ ls -li original.txt hardlink.txt symlink.txt
11654 -rw-r--r-- 2 himasree himasree 59 Oct 1 09:40 hardlink.txt
11654 -rw-r--r-- 2 himasree himasree 59 Oct 1 09:40 original.txt
11656 lrwxrwxrwx 1 himasree himasree 12 Oct 1 09:41 symlink.txt -> original.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ ls -li original.txt hardlink.txt symlink.txt
11654 -rw-r--r-- 2 himasree himasree 59 Oct 1 09:40 hardlink.txt
11654 -rw-r--r-- 2 himasree himasree 59 Oct 1 09:40 original.txt
11656 lrwxrwxrwx 1 himasree himasree 12 Oct 1 09:41 symlink.txt -> original.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ stat original.txt
  File: original.txt
  Size: 59          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 11654         Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/himasree)   Gid: ( 1000/himasree)
Access: 2026-10-01 09:40:58.925496204 +0000
Modify: 2026-10-01 09:40:51.907279595 +0000
Change: 2026-10-01 09:40:51.907279595 +0000
 Birth: 2026-10-01 09:40:15.051903866 +0000
himasree@LAPTOP-41COT5AA:~/OSLab$ stat symlink.txt
  File: symlink.txt -> original.txt
  Size: 12          Blocks: 0          IO Block: 4096   symbolic link
Device: 8,48      Inode: 11656         Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/himasree)   Gid: ( 1000/himasree)
Access: 2026-10-01 09:41:15.279217418 +0000
Modify: 2026-10-01 09:41:06.925459872 +0000
Change: 2026-10-01 09:41:06.925459872 +0000
 Birth: 2026-10-01 09:41:06.925459872 +0000
himasree@LAPTOP-41COT5AA:~/OSLab$ stat -L symlink.txt
  File: symlink.txt
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Size: 59          Blocks: 8          IO Block: 4096   regular file
Device: 8,48     Inode: 11654        Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/himasree)   Gid: ( 1000/himasree)
Access: 2026-10-01 09:40:58.925496204 +0000
Modify: 2026-10-01 09:40:51.907279595 +0000
Change: 2026-10-01 09:40:51.907279595 +0000
Birth: 2026-10-01 09:40:15.051903866 +0000
himasree@LAPTOP-41COT5AA:~/OSLab$ find . -inum $(ls -li original.txt | awk '{print $1}')
./hardlink.txt
./original.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ rm original.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ cat hardlink.txt
This is my inode experiment file.
Added through hard link.
himasree@LAPTOP-41COT5AA:~/OSLab$ ls -li hardlink.txt symlink.txt
11654 -rw-r--r-- 1 himasree himasree 59 Oct  1 09:40 hardlink.txt
11656 lrwxrwxrwx 1 himasree himasree 12 Oct  1 09:41 symlink.txt -> original.txt
himasree@LAPTOP-41COT5AA:~/OSLab$ |

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himasree@LAPTOP-41COT5AA:~/OSLab$ nano mmap_demo.c
himasree@LAPTOP-41COT5AA:~/OSLab$ cat mmap_demo.c
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <string.h>

int main() {

    const char *filename = "mmap_file.txt";

    /* Open file */
    int fd = open(filename, O_RDWR | O_CREAT, 0644);

    if (fd == -1) {
        perror("open");
        return 1;
    }

    /* Size required for the file */
    const char *text =
        "Hello from mmap!\n"
        "This file is accessed using memory mapping.\n";

    size_t size = strlen(text);

    /* Set file size */

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    if (ftruncate(fd, size) == -1) {
        perror("ftruncate");
        close(fd);
        return 1;
    }

    /* Map file into memory */
    char *mapped = mmap(
        NULL,
        size,
        PROT_READ | PROT_WRITE,
        MAP_SHARED,
        fd,
        0
    );

    if (mapped == MAP_FAILED) {
        perror("mmap");
        close(fd);
        return 1;
    }

    printf("File successfully mapped into memory.\n");

    /* Write through memory mapping */
    memcpy(mapped, text, size);

    printf("\nData written using mmap():\n");
    printf("%.s", (int)size, mapped);

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printf("File successfully mapped into memory.\n");

/* Write through memory mapping */
memcpy(mapped, text, size);

printf("\nData written using mmap():\n");
printf("%.s", (int)size, mapped);

/* Modify mapped memory */
mapped[0] = 'h';

/* Ensure changes are written to file */
if (msync(mapped, size, MS_SYNC) == -1) {
    perror("msync");
}

printf("\nAfter modifying mapped memory:\n");
printf("%.s", (int)size, mapped);

/* Unmap */
if (munmap(mapped, size) == -1) {
    perror("munmap");
}

close(fd);

printf("\nMemory mapping removed.\n");
printf("File closed successfully.\n");

```

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himasree@LAPTOP-41COT5AA:~/OSLab$ gcc -Wall -g mmap_demo.c -o mmap_demo
himasree@LAPTOP-41COT5AA:~/OSLab$ ./mmap_demo
File successfully mapped into memory.

Data written using mmap():
Hello from mmap!
This file is accessed using memory mapping.

After modifying mapped memory:
hello from mmap!
This file is accessed using memory mapping.

Memory mapping removed.
File closed successfully.
himasree@LAPTOP-41COT5AA:~/OSLab$ |
```

```
himasree@LAPTOP-41COT5AA:~/OSLab$ cat mmap_file.txt
hello from mmap!
This file is accessed using memory mapping.
himasree@LAPTOP-41COT5AA:~/OSLab$ |
```